Contextual Provenance Defense for Persistent Memory Poisoning in LLM Agents

Literature Review and Brief Project Proposal

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1 Introduction

LLM agents use memory to keep preferences, facts and earlier task outcomes. This helps them continue work across sessions, but it also gives an attack somewhere to stay. An instruction hidden in an email or tool response can be saved and affect a later decision. Our project focuses on detecting this persistent memory poisoning without blocking useful memory.

Review scope

This focused review compares ten studies from 2023-2026. Nine appeared at ACM UIST, ACL, NeurIPS, ICML or ICLR; MemGPT is an arXiv preprint. Three additional preprints provide recent security evidence. References and reported figures were checked against original paper or venue records. Limitations below include our assessment of how each method fits this project.

2 Literature Review

2.1 How agents organise memory

MemGPT uses separate memory tiers and moves information in and out of the context window [1]. Generative Agents stores experiences, retrieves relevant ones and turns them into reflections for planning [2]. A-Mem goes further by linking notes and updating older memories when new information arrives [3]. These approaches improve continuity and reasoning. Their main evaluations, however, focus on useful recall or believable behaviour. They do not establish that stored information is safe. Linking and summarising memory also creates more places where an untrusted claim could spread.

2.2 Measuring useful memory

LoCoMo tests question answering, summarisation and dialogue over long conversations [4]. LongMemEval tests extraction, reasoning across sessions, time, updates and abstention using 500 questions [5]. Both show that memory remains difficult even without an attacker. LoCoMo reports that long-context and retrieval methods still lag behind humans. LongMemEval reports about a 30% accuracy drop across sustained interactions. These benchmarks matter for a defense: deleting every suspicious record may improve security while making the assistant forget information it needs.

2.3 Attacker access makes a difference

AgentPoison assumes the attacker can insert poisoned demonstrations into a memory or knowledge store [6]. Its optimised trigger achieves 81.2% average retrieval success with under 0.1% poisoning and a small change in benign accuracy. MINJA removes direct store access: the attacker uses normal queries to make the agent produce and save malicious reasoning [7]. It reports 98.2% injection success and 76.8% success in eliciting the targeted reasoning. The figures measure different outcomes and should not be ranked directly. Together, they show why testing clean accuracy alone can miss a poisoned agent.

2.4 Comparing defense approaches

PIGuard is a DeBERTa classifier designed to detect prompt injection without rejecting benign text just because it contains suspicious words [8]. Its NotInject dataset contains 339 benign examples. The paper reports 83.48% average accuracy and 87.32% accuracy on the over-defense test. It is a useful low-cost baseline, but its input does not directly model a record's relationship with a later action or other stored memories.

A-MemGuard addresses that contextual problem [9]. It compares reasoning paths based on related memories and keeps detected failures in a separate lesson memory. In one reported EHRAgent setting, retrieval attack success falls from 100% to 2.13%. This is a retrieval result, rather than proof that all harmful actions disappear. Its repeated reasoning paths add cost, and coordinated poisoning could undermine the majority view used for validation.

BlindGuard learns normal patterns in multi-agent communication using a hierarchical encoder and contrastive corruption [10]. It detects unfamiliar attacks without needing labels for every attack type. Its evaluation includes prompt, tool and memory attacks on different communication graphs. The approach supports using relationships as evidence, but its nodes are agents. Applying it to stored memories would require an adaptation and a fresh evaluation.

2.5 Recent evidence on persistence and authority

MPBench studies memory poisoning in OpenClaw and HERMES, including inputs such as documents, email and websites [11]. It reports 50.46% average attack success and 41.05% retrieval success. MemSecBench follows 310 cases through writing, execution and repair across 24 configurations [12]. Malicious memory persists in 84.2% of cases, while the full write-execute chain succeeds in 50.3%. This makes the distinction between storing poison and causing harm particularly clear.

TMA-NM takes a different approach: it binds a record to its origin and limits the authority it can gain through summaries or tool echoes [13]. The authors report 0% attack success within their tested direct and laundering settings. That result depends on the stated trust assumptions; it is not a guarantee against a compromised host. These three studies are preprints, so their findings are treated as provisional supporting evidence.

2.6 What the studies show together

Memory architectures and clean benchmarks explain what the agent needs to preserve. Attack studies show how that same memory can redirect later reasoning. The defenses then trade off different costs: PIGuard checks text cheaply, A-MemGuard uses repeated contextual reasoning, and BlindGuard uses interaction structure. Origin-based controls address permission to act. We therefore need to measure harmful outcomes alongside useful recall and latency. A defense that recognises an attack but still lets the agent act on it has only solved part of the problem.

3 Comparison of the Selected Studies

Table 1 Problems, methods, data, results and limitations

Study	Problem	Method / approach	Dataset / data	Key results	Limitations
MemGPT [1] 2023 Preprint	Limited context for long tasks	LLM-managed working and external memory tiers	Multi-session chat, document QA and nested key-value retrieval	Supports recall beyond the context window and continuity across sessions	No dedicated memory-poisoning evaluation
Generative Agents [2] 2023 ACM UIST	Maintaining coherent simulated behaviour	Memory stream, relevance-based retrieval, reflection and planning	25 agents in a simulated town; human ratings and ablations	Memory and reflection contribute to believable behaviour	Small simulated setting; poisoning resistance not established
A-Mem [3] 2025 NeurIPS	Rigid organisation of stored experience	Linked notes, dynamic indexing and memory evolution	LoCoMo and DialSim; six foundation models	Improved multi-hop reasoning; DialSim F1 3.45 versus 2.55 for its LoCoMo baseline	Utility-focused evaluation; safety of updates and links not established
LoCoMo [4] 2024 ACL	Insufficient evaluation of very long conversations	Human-checked generated dialogues and memory tasks	Conversations averaging 600 turns and 16K tokens, up to 32 sessions	Long-context and RAG methods improve recall but remain below human performance	Constructed dialogue data; no persistent attack evaluation
LongMemEval [5] 2025 ICLR	Weak long-term interactive recall	Five ability tests; indexing, retrieval and reading analysis	500 questions within scalable user-assistant histories	About 30% accuracy drop over sustained interactions; indexing improvements help	Clean-memory benchmark; does not measure poisoning

Results are those reported within each paper's own setting. Retrieval success, targeted reasoning, task accuracy and end-to-end attack success are different measures.

3 Comparison of the Selected Studies continued

Table 1 Continued

Study	Problem	Method / approach	Dataset / data	Key results	Limitations
AgentPoison [6] 2024 NeurIPS	Backdoors in agent memory or knowledge stores	Optimised query trigger retrieves poisoned demonstrations	Driving, StrategyQA and healthcare EHR agents	81.2% mean retrieval success; under 0.1% poisoning; benign accuracy change 0.74%	Requires direct store access and a trigger in later queries
MINJA [7] 2025 NeurIPS	Poisoning without direct store access	Query-only interaction, bridging reasoning and progressive shortening	Healthcare, web and question-answering agents	98.2% injection success; 76.8% targeted-reasoning success	Needs opportunities to influence shared memory and target-specific interactions
PIGuard [8] 2025 ACL	False rejection from suspicious vocabulary	DeBERTa classifier with over-defense mitigation	NotInject and benign/malicious prompt-injection sets	83.48% average accuracy; 87.32% on NotInject; 15.34 ms in its test setting	Pointwise prompt guard; later memory context is not explicitly modelled
A-MemGuard [9] 2026 ICML	Context-dependent poison and error reinforcement	Consensus across reasoning paths; separate lesson memory	StrategyQA, EHRAgent, MMLU and MISINFOTASK settings	Example EHRAgent retrieval ASR: 100% to 2.13%; end-to-end outcomes vary	Extra reasoning cost; majority evidence can be unreliable
BlindGuard [10] 2026 ACL	Unseen attacks spreading across agents	Hierarchical graph encoder; corruption-based contrastive training	CSQA, MMLU and PoisonRAG-based multi-agent tests	Improved unknown-attack transfer; detects attacks without attack-specific training labels	Designed for agent graphs; memory-record adaptation remains untested

Limitations include our assessment of suitability for this project. A-Mem and A-MemGuard are separate methods. The three supplementary preprints [11]-[13] are discussed in Section 2.5.

4 Limitations and Research Gaps

The first gap is the separation between memory usefulness and memory security. MemGPT, Generative Agents and A-Mem show why agents benefit from storing experience, but their main results do not demonstrate poisoning resistance [1]-[3]. LoCoMo and LongMemEval can measure what a guard forgets or wrongly blocks, but need separate attack tests [4], [5].

The second gap concerns context and cost. A-MemGuard already uses contextual validation, so context alone cannot be claimed as our novelty [9]. The question is whether a smaller model can check a record with the current query, related memories and source information while reducing repeated LLM calls. BlindGuard suggests that relationships help, but transfer from agent graphs to memory records still needs testing [10].

A third issue is where success is measured. Poison can be stored without being retrieved, or retrieved without causing the intended harm [6], [12]. Published percentages come from different agents and protocols. A fair comparison needs the same agent configuration and separate measures for writing, retrieval and harmful actions.

Source authority also needs its own control. A document can be factually useful without having permission to approve a payment or change a setting. TMA-NM highlights why source identity should survive summarisation [13]. A learned risk score should not create that permission.

5 Proposed Direction

We propose Contextual Provenance Memory Guard (CPMG), a compact transformer that reads a candidate memory with its source interaction, the current query or planned action, and nearby memories. Source metadata would be recorded by the system rather than accepted from claims inside the text. The guard would check new writes and recheck retrieved records when their later context is available.

MPBench would provide the main attack data [11]. We would add controlled cases where the harmful meaning is split across records or appears only after a later trigger, with matched benign examples. PIGuard and A-MemGuard would be the main baselines. A memory-graph adaptation of BlindGuard would be an additional comparison, clearly labelled as an adaptation [8]-[10].

Evaluation would report end-to-end attack success, false rejection, latency and useful recall on LongMemEval [5]. Scenario groups would be kept separate across training and testing, and at least one attack family would be held out. Removing context or source features would test whether they actually help. Protected tool actions would also require a separate authority check. Our expected contribution is evidence about this security-cost trade-off; no performance improvement is assumed in advance.

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